

TRIASHA SARKAR

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Machine learning engineer and Georgia Tech aerospace MS graduate who builds tested systems from data and evaluation through deployment. Experience spans search and ranking, recommendation, GenAI evaluation, time-series modeling, scientific ML, and safety-critical engineering.

EDUCATION

MS, Aerospace Engineering Georgia Institute of Technology, Atlanta, GA <i>Graduate Research Assistant, Aerospace Systems Design Laboratory (ASDL)</i>	Aug 2026
MSc, Aerospace Vehicle Design Cranfield University, United Kingdom <i>First Class Honours · Thesis on geometry optimization under FAA constraints</i>	Feb 2023
B.Tech, Aerospace Engineering SRM Institute of Science and Technology, India <i>First Class Honours</i>	Jun 2021

TECHNICAL SKILLS

ML & GenAI: PyTorch, scikit-learn, Transformers, RAG, hybrid retrieval, reranking, recommender systems, uncertainty estimation, time-series ML
Systems: Python, SQL, FastAPI, Docker, Kubernetes, GitHub Actions, DuckDB, PostgreSQL/pgvector, ONNX, Core ML, Qualcomm QNN
Scientific computing: NumPy, SciPy, pandas, NetworkX, OSMnx, statistical inference, design of experiments, surrogate modeling

PROFESSIONAL EXPERIENCE

Graduate Research Assistant under Prof. Dimitri Mavris Georgia Tech ASDL - Faculty and sponsors were making decisions from aerospace simulation and safety studies. I audited the data, built the models, and tested each result before delivery. - I checked assumptions with domain experts and documented the limits, which made the analyses easier to review and reproduce.	May 2025 – Aug 2026
Machine Learning Engineer ALTEN India, embedded at Rolls-Royce - Mixed-frequency engine data made early fault signals hard to see. I built Python pipelines for diagnostics, anomaly detection, and predictive maintenance, then reviewed each output with Rolls-Royce domain experts. - Certification work required a traceable link from requirement to result, so I turned engineering rules into repeatable analyses and model checks.	Oct 2023 – Apr 2025
Data Science Intern Rolls-Royce DataLabs Recurring failure patterns were buried in aircraft-engine sensor data. I cleaned the records, designed features, and compared predictive and anomaly-detection models for engineering review.	May – Jul 2021

TECHNICAL PROJECTS

AeroRAG-X Retrieval and LLM Evaluation for Technical Knowledge <i>github.com/triasha72/AeroRAG-X</i> - NASA reports were hard to search and verify, so I indexed 3,233 source-preserving chunks and combined hybrid retrieval, reranking, pgvector, and citation controls. - The system had to refuse unsupported answers before they reached a user. I added protected tests, evidence gates, bounded agents, and traced FastAPI services; a separate release check blocks placeholder GRPO runs.	2025 – Present
NewsLens Recommendation, Ranking, and Production ML <i>github.com/triasha72/NewsLens</i> - Future clicks can inflate recommender scores. I prevented that leakage with chronological splits, personalized TF-IDF ranking, and a training-only fallback, reaching 0.366 NDCG@10. - After freezing the design, I ran the official holdout once; the gain was positive but its interval crossed zero, so I kept the original baseline. I deployed it with DuckDB, FastAPI, Docker, and two local Kubernetes replicas.	Jul 2026
EdgeGenBench Scientific ML and On-Device Inference <i>github.com/triasha72/EdgeGenBench</i> - Fast aircraft trade studies still had to respect physics checks. I built a surrogate benchmark with uncertainty estimates, constrained optimization, ONNX export, and deployment tests. - The Qualcomm model kept 0.997 held-out R^2, and the deployment policy rejected a drifting INT8 candidate. I also shipped an installable Safari app as the browser-based iPhone path.	Aug 2026 – Present
AeroSynth-Eval Multimodal AI Evaluation <i>github.com/triasha72/AeroSynth-Eval</i> - The VLM benchmark could not shift between experiments. I froze 48 scenarios with deterministic images, provenance checks, annotation workflows, and typed evaluator contracts. - On a Kaggle P100, 8 of 12 development cases met the response schema. I kept the four JSON failures to guide the next fix instead of reporting them as successes.	Aug 2026 – Present

Silent Failure Detection in Rocket-Motor Simulations

May – Jul 2026

Georgia Tech ASDL · Graduate Research Assistant under Prof. Dimitri Mavris

- A design sweep was hanging without an error, so I audited the data and source code, trained a leakage-safe classifier with 96.8% accuracy, and traced the fault to an uncapped convergence loop.
- I confirmed the failure pattern on unseen geometries and an independent sample, then patched the loop to turn silent hangs into logged convergence errors.

HERO | Source-Aware Retrieval for Airline Safety

2025 – 2026

Georgia Tech ASDL · Graduate Research Assistant under Prof. Dimitri Mavris

- Delta safety analysts had to trace generated answers back to evidence. I developed retrieval and evaluation methods that keep each answer tied to the technical sources used.

GREEN TEA and Project EAGLE | Sustainable Aviation Modeling

Sep 2025 – Apr 2026

Georgia Tech ASDL · Graduate Research Assistant under Prof. Dimitri Mavris

- Alternative-fuel studies were too slow for repeated sponsor trade-offs. I built surrogate, uncertainty, demand, and life-cycle models, and the GREEN TEA model remains in sponsor use.

Atlanta Mobility Resilience Digital Twin

2026 – Present

github.com/triasha72/atlanta-mobility-resilience-digital-twin

- Road-closure studies needed a repeatable way to measure travel time and access. I built an OSMnx and NetworkX simulator with deterministic tests in CI; live map download remains a separate integration check.

Surrogate Model Learning

2025 – Present

github.com/triasha72/Surrogate-model-learning

- Common surrogate methods fail in different parts of the design space. I compared Gaussian process, response-surface, and radial-basis models under the same experiments, while clean-kernel CI keeps the notebooks reproducible.

Equity Backtest | Walk-Forward Signal Evaluation

Jul 2026

github.com/triasha72/Equity-Backtest

- A momentum signal only matters if it survives time ordering and costs. I built an expanding-window backtest and logged every specification; the apparent edge disappeared after costs and missed the pre-set significance threshold.

LEADERSHIP

Graduate Chair | Women of Aeronautics and Astronautics, Georgia Tech

2025 – 2026